disputes, but to prevent disputes from arising."[13] The objective textual approach prioritizes clear, simple, and precise resolution of interpretative dispute. As a result, subjective intent articulated in the preparatory work and circumstances of conclusion is subjugated as a supplementary means of interpretation.

For the purposes of the analysis in this article, the Vienna Convention is adopted as the methodological approach of treaty interpretation subject to the following caveat. While the Convention does provide a reasonable method of interpretation, the Outer Space Treaty warrants Article 32 of the Convention to play a more prominent role than perhaps it would otherwise when interpreting other treaties.

With the exception of Article IV of the Outer Space Treaty and provisions related to procedural aspects of it, all other Treaty articles are articulations, and is some cases verbatim copies, of the principles articulated in GA Res 1962 (XVIII), *Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space*.[14] As a result, the Outer Space Treaty is primarily a treaty of principles, crafted for the purposes of *proscribing* norms to an area that was *without law*. These proscriptive principles, by their very nature, cannot embody their object and purpose by solely reviewing their text. It is for this reason that the historical circumstance of human advancement into outer space is closely intertwined with the object and purpose of these principles.

Article 32 of the Vienna Convention does allow for supplementary means of interpretation, including the preparatory work of the treaty and the circumstances of its conclusion, in order to confirm the meaning resulting from the application of Article 31. When interpreting the Outer Space Treaty, confirmation of the meaning of text should be undertaken with an examination of the preparatory work of the treaty and circumstances of its conclusions.

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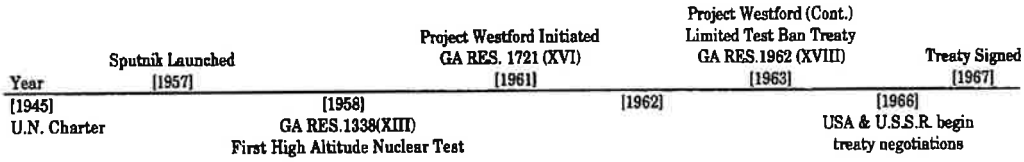
[13] Vandervelde, *supra* note 9 at 342.

[14] Declaration of Legal Principles Governing the Activities of States in the Exploration and Use of Outer Space, GA Res. 1962(XVIII) [hereinafter Declaration].

III. HISTORICAL CONTEXT OF ARTICLE IX

Prior to assessing Article IX obligations, it is necessary to have an understanding of the historical process that led to Article IX and the Outer Space Treaty. Therefore let us examine, in a chronological order, the historical development of the Article.

Important Historical Events related to Article IX of the Outer Space Treaty



A. *U.N. Charter, Sputnik, HANDs, and Project West Ford*

The UN Charter entered into force on October 24, 1945. Article 13(a) of the Charter grants to the General Assembly authority to initiate studies and make recommendations for the purpose of "promoting international co-operation in the political field and encouraging the progressive development of international law and its codification."[15]

On October 4, 1957, the former Soviet Union launched *Sputnik,* the first manmade satellite, into orbit. The launch of *Sputnik* accentuated the debate on what law, if any, does apply to outer space, and on what law, if any, should apply. The United Nations General Assembly took action on December 13, 1958, and in accordance with their authority under Article 13(a) of the Charter, passed GA Res. 1348(XIII).[16] This resolution established the ad hoc Committee on Peaceful Uses of Outer Space (COPUOS) and tasked the Committee to report to the General Assembly on "the nature of legal problems which may arise in the carrying out of programs to explore outer space."[17]

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[15] U.N. Charter art 13(a).

[16] Question of the Peaceful Use of Outer Space, GA Res. 1348 (XIII) (1958).

[17] *Id.* at para. (1)(d).

COPUOS was established as a full committee on December 12, 1959.[18]

1958 was also the first year of high altitude nuclear detonations (HAND). From 1958 until 1962 the United States and Soviet Union conducted a series of HANDs. These nuclear tests affect the operation of applications that utilize outer space and the Earth's atmosphere. The COPOUS Scientific and Technical Sub-Committee considered high altitude and outer space nuclear weapon tests one aspect of potentially harmful space interference that needed to be addressed.[19]

In 1961 the General Assembly articulated two principles that were later incorporated into the Outer Space Treaty: "International law, including the Charter of the United Nations, applies to outer space and celestial bodies" and "outer space and celestial bodies are free for exploration and use by all States in conformity with international law and are not subject to national appropriation."[20]

Also in 1961 the United States initiated *Project West Ford*, a space communications experiment. *Project West Ford* involved placing hundreds of millions of small copper dipoles into orbit[21] for the purpose of investigating the technical feasibility of using such dipoles as passive reflectors for communications.[22] *Project West Ford* raised concerns amongst the international community that the copper dipoles could cause potentially harmful inference with radio astronomy, optical astronomy, space communications, and space travel.[23] The International Astronomical Union (IAU) passed a resolution appealing "to all governments . . . launching space experiments which could possibly affect as-

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[18] International Cooperation in the Peaceful Uses of Outer Space, GA Res. 1472(XIV) (1959).

[19] Opening Statement by the Chairman, GA 18th Sess., Annex II, para. (e). UN Doc. A/5482 (1963) at 10. *See also*, *The Peaceful Uses of Outer Space*, *in* 1962 UNITED NATIONS YEARBOOK 42 (1962) [hereinafter UNITED NATIONS YEARBOOK].

[20] International Cooperation in the Peaceful Uses of Outer Space, GA Res. 1721(XVI) (1961).

[21] *See* C.F.J. Overhage & W.H. Radford, *The Lincoln Laboratory West Ford Project – An Historical Perspective*, *in* 52 (5) PROC. OF THE IEEE 452-54 (May 1964) [hereinafter *The Lincoln Laboratory*].

[22] KARL-HEINZ BOCKSTIEGEL ET AL., *United States Space Communication Experiment*, *in* SPACE LAW BASIC LEGAL DOCUMENTS, at A.IX.1.2 (2007).

[23] *See The Lincoln Laboratory*, *supra* note 21.

tronomical research to consult with the IAU before undertaking such experiments and to refrain from launching until it is established beyond doubt that no damage will be done to astronautical research."[24] The International Council of Scientific Unions (ICSU) Committee on Space Research (COSPAR) established the Consultative Group of Potentially Harmful Effects of Space Experiments (CGPHESE). CGPHESE conducted a study of *Project West Ford* and recommended that "any proposals for future experiments of this sort" should be evaluated thoroughly by the scientific community.[25] It was within this historical context that in 1962 the Soviet Union submitted the first draft declaration in COPOUS to address the issue of harmful interference.

B. COPUOS and General Assembly Resolution 1962(XVIII)

GA Res. 1962(XVIII), the progenitor to the Outer Space Treaty, was drafted in 1962 and 1963. This drafting process involved members of COPUOS circulating a series of draft declarations and negotiating amongst themselves to achieve consensus on a declaration that could be transmitted to the First Committee of the General Assembly.

On June 6, 1962, the Soviet Union submitted a draft declaration to be considered by COPUOS (U.S.S.R. Draft). The U.S.S.R. Draft contained a provision that stated in paragraph 6: "Co-operation and mutual assistance in the conquest of outer space shall be a duty incumbent upon all States; the implementation of any measures that might in any way hinder the exploration or use of outer space for peaceful purposes by other countries shall be permitted only after prior discussion of and agreement upon such measures between countries concerned."[26]

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[24] DELBERT TERRILL, THE AIR FORCE ROLE IN DEVELOPING INTERNATIONAL OUTER SPACE LAW 65 (Maxwell Air Force Base: Air University Press, 1999) (quoting R. Cargill Hall, The International Legal Problems in Space Exploration, An Analytical Review 41-42 (June 1966) (unpublished master's thesis, California State University at San Jose)).

[25] Z. Niemirowicz, *A Marriage to COSPAR: Part I - 1957-1967*, 1993 COSPAR INFORMATION BULLETIN 77-88.

[26] U.S.S.R. Draft Declaration of the Basic Principles Governing the Activities of States in the Exploration and Use of Outer Space, GA 18th Sess., Annex III, para.6. UN Doc. A/5482 at 11 (1963) [hereinafter Draft Declaration].

The U.S.S.R. Draft met resistance from the U.K. and the United States.[27] The U.K. and United States maintained that the U.S.S.R. Draft "introduced a veto on the activities of other in outer space" and seemed less effective than action already taken by COSPAR, which had established CGPHESE.[28] France "agreed with the idea of consultations between States engaged in the exploration of outer space in order not to obstruct outer space activities of other States" but "objected to making prior consent imperative for the such activities by another State."[29]

On December 4, 1962, the U.K. submitted to COPUOS a draft declaration (U.K. Draft) that responded to the Soviet position. The U.K. Draft contained a provision that stated exploration and use of outer space "shall be exercised by all States with due regard to the interests of other States in the exploration and use of outer space, and to the need for consultation and co-operation between States in relation to such exploration and use."[30] This draft differed from the U.S.S.R. Draft in two significant ways. First, while the U.S.S.R. Draft spoke of mutual assistance and co-operation as a duty, the U.K. Draft articulates the principle of due regard. Second, the U.K. draft did not require prior consent, nor did it place an affirmative duty to consult. Instead the U.K. draft simply spoke of the need for consultation and co-operation.

Ultimately COPUOS adopted a provision that struck a balance between the U.S.S.R. draft and the U.K. draft, similar to the position articulated by France. Paragraph 6 of *Declaration of Legal Principles Governing the Activities of Outer Space* (Declaration) states:

> In the exploration and use of outer space, States shall be guided by the principle of co-cooperation and mutual assistance and shall conduct all their activities in outer space with due regard for the corresponding interests of other States. If a State has reason to believe that an outer space activity or experiment planned by it or its nationals would cause potentially

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[27] UNITED NATIONS YEARBOOK, *supra* note 19, at 42.
[28] *Id.*
[29] *Id.*
[30] Draft Declaratión, *supra* note 26, at 11.

> harmful interference with activities of other States in the peaceful exploration and use of outer space, it shall undertaken appropriate international consultations before proceeding with any such activity or experiment. A State which has reason to believe that an outer space activity or experiment planned by another State would cause potentially harmful interference with activities in the peaceful exploration and use of outer space may request consultation concerning the activity or experiment.[31]

As will be discussed in further detail below, the language of the Declaration imposes an affirmative duty to consult but does not require prior consent. The Declaration was adopted by the General Assembly on December 13, 1963, as GA Res. 1962(XVIII).

Comparison of Draft Declarations and the Evolution of Paragraph 6 GA Res. 1962 (XVIII)

U.S.S.R. Draft Declaration June 6th, 1962	U.K. Draft Declaration December 4th, 1962	Draft Declaration accepted by consensus and adopted as G.A. Res. 1962 (XVIII)
• Prior consent required for "any measures that might in any way hinder the exploration and use of outer space" • Duty of "cooperation and mutual assistance"	• Due regard to the interests of other States in the exploration and use of outer space • The "need for consultation and cooperation between States in relation to such exploration and use"	• Due regard for the corresponding interests of other States • Appropriate international consultations required for activities planned that State has reason to belcive "would cause potentially harmful interference with activities of other States in the peaceful exploration and use of outer space"

[31] Declaration, *supra* note 14, at para. 6.

C. Limited Test Ban Treaty

The Treaty Banning Nuclear Weapon Tests in the Atmosphere, in Outer Space and Under Water (Limited Test Ban Treaty) entered into force on October 10, 1963. This treaty addressed one issue of potentially harmful interference—the detonation of nuclear devices in the atmosphere and in outer space. Each Party to the the Limited Test Ban Treaty undertakes "to prohibit, to prevent, and not to carry out nuclear wepaon test exploration, or any other nuclear exposion, at any place under its jurisidction or control" in the atmosphere and beyond its limits, including outer space.[32]

D. Outer Space Treaty and Article IX

The Outer Space Treaty was negotiated in COPUOS in 1966 and entered into force October 10, 1967. With the exception of the harmful contamination clause, the operative language of Article IX is a verbatim copy of GA Res. 1962 (XVIII) paragraph 6.[33] The harmful contamination provision was proferred by Canada during the 1962-1963 COPUOS negotiations on the *Declaration of Principles,* but was not included in the final draft Declaration.[34] Canada voiced concern that with the exclusion of a harmful contamination provision from the *Declaration* a Party "is not specifically asked to undertake consultation if an experiment planned by it or its nationals might involve a risk of modifying the natural environment of the Earth in a manner likely to be prejudicial for the well-being of human life or the interests of another State."[35]

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[32] Treaty Banning Nuclear Weapon Tests in Atmosphere in Outer Space and Under Water, Oct. 10, 1963, 480 U.N.T.S. 43.

[33] Article IX of the Outer Space Treaty added additional language to formulate ¶ 6 of GA Res.1962(XVI) to ensure enforceability as a treaty provision ("State Party to the Treaty") and to clarify the scope of outer space to include the moon and other celestial bodies. *See* Outer Space Treaty, *supra* note 7, at art. IX.

[34] Verbatim Record of the 24th meeting of COPUOS, Nov. 22, 1963, Doc. A/5549/Add.1 Annex 1 at 17.

[35] *Id.*

During the course of the 1966 Outer Space Treaty negotiations, a harmful contamination provision was added and Article IX of the Treaty was adopted as follows:

> In the exploration and use of outer space In the exploration and use of outer space, including the Moon and other celestial bodies, States Parties to the Treaty shall be guided by the principle of co-operation and mutual assistance and shall conduct all their activities in outer space, including the Moon and other celestial bodies, with due regard to the corresponding interests of all other States Parties to the Treaty. States Parties to the Treaty shall pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose. If a State Party to the Treaty has reason to believe that an activity or experiment planned by it or its nationals in outer space, including the Moon and other celestial bodies, would cause potentially harmful interference with activities of other States Parties in the peaceful exploration and use of outer space, including the Moon and other celestial bodies, it shall undertake appropriate international consultations before proceeding with any such activity or experiment. A State Party to the Treaty which has reason to believe that an activity or experiment planned by another State Party in outer space, including the Moon and other celestial bodies, would cause potentially harmful interference with activities in the peaceful exploration and use of outer space, including the Moon and other celestial bodies, may request consultation concerning the activity or experiment.

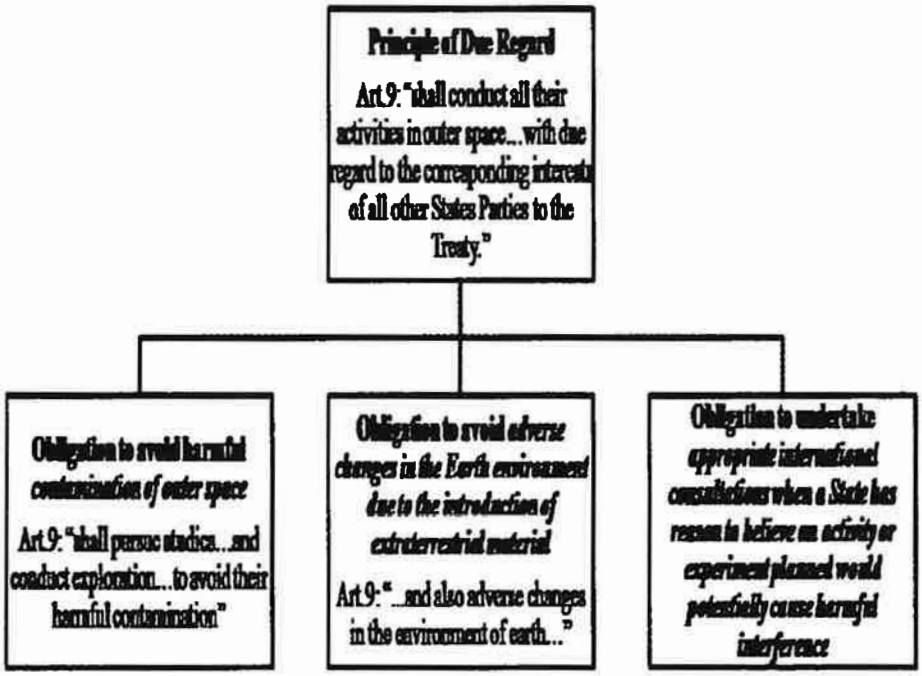
IV. LEGAL INTERPRETIVE ANALYSIS OF ARTICLE IX OBLIGATIONS

Activities in outer space are governed by the principle of due regard as articulated in the first sentence of Article IX, which states: "In the exploration and use of outer space, including the Moon and other celestial bodies, States Parties to the Treaty...shall conduct all their activities in outer space, including the Moon and other celestial bodies, with due regard to the

corresponding interests of all other States Parties to the Treaty."

The second and third sentences of Article IX elaborate upon the principle of due regard and prescribe three *proscriptive* positive legal obligations. First, studies and exploration are to be conducted so as to avoid harmful contamination of outer space. Second, studies and exploration are to be conducted so as to avoid adverse changes in the environment of Earth from the introduction of extraterrestrial matter. Third, appropriate international consultations shall be undertaken before proceeding with any activity or experiment that a State has reason to believe would cause potentially harmfully interference with activities of other States in the peaceful exploration and use of outer space. The matrix below illustrates the relationship of these obligations.

Matrix of Article IX Positive Obligations



It is important to note that Article IX does not distinguish between military and civilian activities, therefore the requirements of Article IX apply fully to military activities in space.[36] The application of Article IX to military activities is however subject to the Charter of the United Nations and general international law, including international law governing armed conflict.[37] As such, in certain situations Article IX obligations may be preempted by other norms of international law.[38]

A. *Obligation to Undertake Appropriate International Consultations*

Article IX of the Outer Space Treaty contains a mandatory international consultation clause that states:

> If a State Party to the Treaty has reason to believe that an activity or experiment planned by it or its nationals in outer space, including the Moon and other celestial bodies, would cause potentially harmful interference with activities of other States Parties in the peaceful exploration and use of outer space, including the Moon and other celestial bodies, it shall undertake appropriate international consultations before proceeding with any such activity or experiment.[39]

In order for a State to be under an obligation to undertake international consultations, three conditions must be satisfied:

> 1) There is an activity or experiment in outer space (e.g. an ASAT activity or test), including the Moon and other celestial bodies, planned by the State or its nationals;[40] and,

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[36] Robert Ramey, *Armed Conflict on the Final Frontier: The Law of War in Space*, 48 A.F. L. REV. 1, 76-77 (2000).

[37] *See* Outer Space Treaty, *supra* note 7, at art. III.

[38] For example, ASAT activities or experiments conducted during a time of armed conflict sanctioned under international law, directed against a belligerent, or sanctioned by the U.N. Security Council necessary for the maintenance of international peace and security may be governed by norms of international law that preempt Article IX positive obligations. *See* Ramey, *supra* note 36. *See also* Michel Bourbonniere, *National-Security Law in Outer Space: The Interface of Exploration and Security*, 70 J. AIR L. & COM. 3, 7-14 (commenting that "the Outer Space Treaty was not meant to change the law governing means and methods of warfare.").

[39] *See* Outer Space Treaty, *supra* note 7, at art. IX.

[40] *Id.*

2) The State must have reason to believe the activity or experiment (e.g. an ASAT activity or test) would cause potentially harmful interference;[41] and,

3) That this potentially harmful interference must potentially interfere with the activities of other States Parties to the Outer Space Treaty in the peaceful exploration and use of outer space, including the Moon and other celestial bodies.[42]

i. Condition 1: Activity or Experiment in Outer Space

The terms "activity," "experiment," "outer space," and "planned" are not defined by the Treaty. The term activity is more encompassing than experiment, as an experiment is only one type of activity that can be undertaken. Thus, except for actions excluded from the scope of Article IX by preemptive norms of international law, the term activity can be reasonably interpreted as any action.

Outer space is not defined under international law and some dispute may arise as to whether the spatial location of an activity is occurring within outer space or airspace.[43] Also, there is a question as to whether or not an activity or experiment that is terrestrial based is also within the scope of the term "in outer space." For example, is an ASAT experiment that targets ground based satellite uplinks to disrupt the operation of orbiting satellites an experiment that is occurring in outer space or is it simply a terrestrial experiment that impacts an object in outer space?

"Planning" for something incorporates an element of premeditation and intent. An unplanned activity or experiment in outer space is possible, although highly unlikely.[44] If an unplanned activity or experiment did occur, the international con-

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[41] *Id.*

[42] *Id.*

[43] For example, a high-altitude activity or experiment may be protested against by a subjacent State on the grounds that the activity or experiment is occurring within the State's sovereign airspace even though no legal delimitation of air and space is established under international law.

[44] As humankind increases outer space utilization, the likely of accidental (e.g. unplanned) outer space activities or experiment will likewise increase, with some level of correlation proportional to use.

sultation clause would not apply. This is a reasonable interpretation because international consultations are required before proceeding with an activity or experiment and one cannot undertake consultations for an activity or experiment they did not plan or intend to conduct.

ii. Condition 2: Reason to Believe that the Activity or Experiment would cause Potentially Harmful Interference

The terms "has reason to believe," "would cause," "potentially," and "harmful interference" are also not defined in the Treaty. "Reason to believe," when read in conjunction with "would cause potentially harmful interference," is indicative of a burden of proof threshold. Reason to believe is not synonymous with certainty and one can exclude certainty of potentially harmful interference as the appropriate interpretation of this provision. "To believe," in this context, is related to holding an opinion or thought.[45] "Reason," when read in conjunction with "to believe" is commonly understood to be a statement of some fact employed to prove or disprove some assertion, idea, or belief.[46] Therefore, reason to believe should be interpreted as having knowledge that proves the assertion that a planned activity would cause potentially harmful interference.

This language "has reason to believe" raises interesting questions. Is this standard of "reason to believe" a subjective or objective standard? If it is subjective, how does a State determine if it has reason to believe? If it is objective, what body decides? These questions illuminate the *principled* nature of the Treaty and illustrate that Article IX was designed to *guide* and provide *proscriptive* general rules of conduct.

"Would cause" is self-explanatory to the extent that the planned activity would result in potentially harmful interference. The potentiality element of the phrase "potentially harmful interference" is abstruse. As one cannot predict with certainty the results of an action before the action is carried out, attempting to predict whether or not a space activity or experi-

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[45] 1 THE SHORTER OXFORD DICTIONARY,(1993).

[46] *Id.*

ment will cause harmful interference is difficult. At the time the phrase "potentially harmful interference" was negotiated, significant concern existed that the planned, but yet conducted, second *Project West Ford* experiment would result in harmful interference to space activities. Furthermore, *Project West Ford*'s purpose was to discover what result, be it interference or otherwise, the dispersal of copper dipoles would have on radio communications. In this sense, the term "potentially" expands the reading of the provision beyond planned actions or experiments that *would* cause harmful interference; and instead encompasses activities and experiments that would cause interference that is potentially harmful.

This in turn leads to the question of what is "harmful interference." Harmful is ordinarily defined as "of a kind likely to be injurious."[47] "Interference" is ordinarily defined as an obstruction or hindrance.[48]

Harmful interference in outer space can be divided into three primary categories: (1) Observational Interference (i.e. either terrestrial based astronomical observations or space based terrestrial observations), (2) Radio Frequency Interference, and (3) Physical Interference (i.e. interference with the freedom of physical movement and/or physical operations in outer space). The ITU defines harmful interference as "interference, which endangers the functioning of a radio navigation service or of other safety services or seriously degrades, obstructs or repeatedly interrupts a radio communication service operating in accordance with Radio Regulations."[49] The ITU definition fits within the category of radio frequency interference.

Read together, the operative language of "has reason to believe that an activity or experiment...would cause potentially harmful interference" places the responsibility and authority to determine whether a State has reason to believe and whether the planned action would cause potentially harmful interference

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[47] MERRIAM-WEBSTER'S ONLINE DICTIONARY, http://www.m-w.com/dictionary/harmful (last visited Oct. 21, 2008).

[48] THE SHORTER OXFORD DICTIONARY, *supra* note 45.

[49] Bourbonniere, *supra* note 38, at 57 (quoting the Convention of the International Telecommunication Union, Dec. 22, 1992, S. Treaty Doc. No. 104-34 (1996)).

with the State charged with the affirmative obligation to consult. As will be discussed in further detail *below*, this in turn allows States a wide degree latitude to determine whether or not this triggering condition is met.

iii. Condition 3: Potentially Harmful Interference with the Peaceful Exploration and Use of Outer Space

Condition 3 requires potentially harmful interference to interfere with the peaceful exploration and use of outer space of other State Parties. This raises the question of whether or not other State Party activities meet the criteria of peaceful use and exploration. If the exploration and use of other States Party to the Treaty are not peaceful, the there is no obligation to undertake appropriate international consultations with regards to potentially harmful interference of non-peaceful use and exploration of outer space. For example, an experiment that would cause potentially harmful interference with a space object of a State Party carrying nuclear weapons in orbit would not trigger Condition 3, so long as the orbiting nuclear weapons are not sanctioned under international law.

iv. What are Appropriate International Consultations?

The Treaty neither proscribes the procedure for appropriate international consultations nor designates an agency to which States should turn for the authoritative evaluation of proposed uses or experiments.[50] As a result, the procedure and substantive nature of "appropriate international consultations" will depend on the nature of the planned activity or experiment.[51] One can logically infer that a State is procedurally obligated, at

[50] Ivan Vlasic, *The Space Treaty: A Preliminary Evaluation*, 55 CAL. L. REV. 507, 517. Also, consider that during the negotiations of GA Res.1962 (XVIII), Australia, Brazil and India suggested that international consultations could be explicitly linked with international forums existing at the time, such as COSPAR's CGPHESE. The United States considered the CGPHESE an appropriate forum for consultation but argued it would be inappropriate to specify one particular mode of conducting international consultations exclusively and for all time. Ultimately, the position of the United States prevailed.

[51] *See* Brandon Hart, *Legal Implications Surrounding Recent Interception of Spy Satellite*, JOINT CENTER FOR OPERATIONAL ANALYSIS (JCOA) J. 34 (June 2008).

minimum, to contact States Parties to the Treaty whose peaceful explorations and use of outer space would experience potentially harmful interference. One can also logically infer that the substantive obligation requires, at minimum, that these States be provided with information sufficient to take appropriate action to prevent potentially harmful interference with their uses or explorations in outer space, the Moon and other celestial bodies.

Consider that the object and purpose of Article IX is guided by principles of "cooperation and mutual assistance" with "due regard to the corresponding interests of all other States Parties to the Treaty."[52] Interpreting the international consultation obligation provision as ad minimum requiring a State to fulfill the aforementioned procedural and substantive obligations is a good faith interpretation of the Treaty given the terms of the Treaty in their context and in the light of its object and purpose. Imposing any less of an obligation would emasculate the international consultation clause of Article IX, a result that is unreasonable.

B. Obligation to Avoid Harmful Contamination

Article IX contains a harmful contamination clause that states: "States Parties to the Treaty shall pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination . . . ."[53]

The Treaty does not define harmful contamination. Harmful is ordinarily defined as "of a kind likely to be injurious."[54] Contamination is ordinarily defined as "to make unfit for use by the introduction of unwholesome or undesirable elements."[55] One can conclude that harmful contamination of outer space is the introduction of elements that make outer space unfit for use or are likely to be injurious to users of outer space.

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[52] *See* Outer Space Treaty, *supra* note 7, at art. IX.
[53] *Id.*
[54] *See* MERRIAM-WEBSTER'S ONLINE DICTIONARY, *supra* note 47.
[55] *Id.*

Article IX does not prohibit harmful contamination of outer space. Instead states are obligated only to *avoid* harmful contamination. The Treaty is silent on appropriate measures or authoritative evaluations to determine whether a State has taken appropriate measures to avoid harmful contamination. Until State practice more clearly establishes appropriate standards for the avoidance of harmful contamination, States are granted a wide degree of latitude to determine what action is appropriate. State action is restricted to the extent that their actions must be conducted in good faith in accordance with the principles of the Treaty and international law.

C. Obligation to Avoid Adverse Changes in the Environment of the Earth

Article IX obligates States to pursue studies and exploration of outer space to avoid "adverse changes in the environment of the Earth resulting from introduction of extraterrestrial matter, and where necessary, shall adopt appropriate measures for this purpose." This provision, in and of itself, is substantively lacking.

The verb "avoid" does not prohibit adverse changes. The adjective "adverse" qualifies the "change" that is to be avoided. Only change resulting from one specific cause, the introduction of extraterrestrial matter, is addressed. A literal interpretation of this provision excludes adverse changes to the Earth's environment resulting from space based electromagnetic radiation sources, the blocking or interference of solar emissions, gravitational and magnetic fields, and other forms of energy that are not considered "extraterrestrial matter."

V. ARTICLE IX ANALYSIS OF *FY-1C* AND *USA-193* ASAT ACTIVITIES

In the following section the *FY-1C* and *USA-193* ASAT activities are assessed in light of Article IX of the Outer Space Treaty. This assessment determines whether or not the obligation for international consultation was applicable and if applicable whether the obligation was fulfilled. Article IX obligations to avoid harmful contamination of outer space are also assessed.

A. FY-1C

On January 11, 2007, China conducted an ASAT test which resulted in the destruction of the *Feng Yun 1C* (*FY-1C*) weather satellite in polar orbit at an altitude of approximately 537 miles. This was the first such destruction of a satellite since the kinetic ASAT missile tests conducted during the Cold War by the United States and the Soviet Union.[56] *Prior* to conducting the *FY-1C* ASAT experiment on January 11, 2007, China did not take any steps to consult or inform the international community. Following the *FY-1C* ASAT test, Britain, Australia, Canada, Japan, Taiwan, India, South Korea and the European Union joined the United States in protesting and calling upon Beijing for consultations.[57] It was not until January 23, 2007, that China publicly confirmed it had conducted the ASAT experiment.[58]

i. International Consultation Obligation

Was China obligated to conduct international consultation, and if so did China satisfy this obligation?

As previously discussed, before a State is obligated to undertake international consultations three conditions must be satisfied: (1) There is an activity or experiment in outer space (e.g. an ASAT activity or test), including the Moon and other celestial bodies, planned by the State or its nationals,[59](2) the State must have reason to believe the activity or experiment (e.g. an ASAT activity or test) would cause potentially harmful interference,[60] and (3) that this potentially harmful interference must potentially interfere with the activities of other States Parties to the Outer Space Treaty in the peaceful exploration

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[56] Kan, *supra* note 4.

[57] Theresa Hitchens, *US-Sino relations in Space: From "War of Words" to Cold War in Space*, 3 CHINA SECURITY 12, 23 (2007).

[58] *China Confirms Satellite Downed,* BBC NEWS REPORT (Jan. 23, 2007), http://news.bbc.co.uk/2/hi/asia-pacific/6289519.stm.

[59] *See* Outer Space Treaty, *supra* note 7, at art. IX.

[60] *Id.*

and use of outer space, including the Moon and other celestial bodies.[61]

On January 11th, 2007, China conducted a kinetic ASAT experiment against the weather satellite, *FY-1C*, operating in a polar orbit. An ASAT test against an orbiting satellite falls within a good faith interpretation of "activity or experiment in outer space."[62] Therefore, the first condition triggering mandatory international consultation is satisfied.

A successful ASAT test against an orbiting satellite with a kinetic kill vehicle will result in the destruction of the targeted satellite and the creation of dangerous fast-moving space debris. The resulting space debris is harmful interference that has the potential to damage or destroy other objects in outer space, in particular objects operating in similar orbits or intersecting orbits. The destruction of *FY-1C* is a large space debris generating event, with thousands of pieces of debris cataloged by the United States Space Surveillance Network.[63] Furthermore, due to the altitude and orbital mechanics of *FY-1C*, this debris threatens all spacecraft flying below 2,000 km and will remain in orbit for decades.[64] As a result, this debris caused and is causing potentially harmful interference with the activities of the International Space Station, the Space Shuttle, and other space objects in polar orbit and LEO which were and are conducting peaceful exploration and use of outer space.[65] This harmful interference can be categorized as physical interference with physical movement and/or operation of vehicles and personnel in outer space.

The Chinese should have had "reason to believe" that the successful destruction of *FY-1C* would create a debris field that would cause potentially harmful interference. The Chinese are

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[61] *Id.*

[62] *Id.*

[63] *Id.* *See also* The Space Security Index 2007 at 6 (Waterlo, Canada: Project Ploughshares, 2007) at 6.
2007).

[64] Frank Morring, *China ASAT Test Called Worst Single Debris Event Ever*, AVIATION WEEK AND SPACE TECHNOLOGY (Feb. 11, 2007), http://www.aviationweek.com /aw/generic/story_generic.jsp?channel=space&id=news/aw021207p2.xml&headline =China%20Asat%20Test%20Called%20Worst%20Single%20Debris%20Event%20Ever>.

[65] *Id.*

sophisticated enough to conduct an ASAT test against an orbiting satellite, are aware of the basic laws of physics that underlies kinetic ASATs and the motion of objects in outer space, and therefore have sufficient knowledge to conclude that their planned activity would create space debris and that resulting debris field would be long lasting and threaten spacecraft over a range of orbits. Of course, it can still be argued that subjectively the Chinese assessed prior to conducting the experiment that there was no reason to believe potentially harmful interference would occur. The weakness of this position is an implied presumption of certainty as to the potentiality of harmful interference. To argue that one had *no reason to believe* that *potentially* harmful interference would occur is beyond the standards of good faith and due regard that are foundation of Article IX.

Interpretation of treaty obligations can be construed either broadly or narrowly. States often interpret agreements to provide greatest latitude of State freedom of action. However, even if Article IX is interpreted broadly for the purpose of providing State freedom of action, the principle of due regard and good faith, when read in conjunction with terms of the Treaty in light of its object and purpose, cannot justify an interpretation that concludes China did not have "reason to believe" not "reason to know," but only the "reason to believe."

It is on these grounds that the Chinese "had reason to believe" and the second condition triggering mandatory international consultation obligation is satisfied.

Other States Party to the Treaty, including the United States, were peacefully using and exploring outer space in polar orbits or intersecting LEO orbits with *FY-1C* on January 11, 2007. The peaceful exploration and uses of space within these orbits are diverse, ranging from the *International Space Station* to weather satellites. On this basis the third condition triggering mandatory international consultations under Article IX is satisfied.

Since these three conditions were met prior to China conducting the *FY-1C* ASAT experiment, China was obligated to conduct appropriate international consultations. These international consultations were obligated to be conducted appropriately with other States Parties to the Treaty that China had

reason to believe the *FY-1C* ASAT experiment would cause potentially harmful interference to their respective peaceful uses and exploration of outer space.

China did not take any steps to consult or even inform any such States Party to the Treaty *prior* to conducting the *FY-1C* ASAT experiment. It was not until January 23, 2007, that China publicly confirmed it had conducted the ASAT experiment.[66] The Treaty grants States broad discretion by not defining appropriate international consultations. However, as discussed above, ad minimum it can be inferred that appropriate international consultations is the transmission of information to potentially affected States that is sufficient for affected States to take appropriate action to prevent potentially harmful interference with their activities or experiments. China took *no action* prior to conducting the *FY-1C* ASAT experiment, not even *informing* potentially affected States of the upcoming activity. China's nonfeasance was a violation of Article IX's obligation to conduct appropriate international consultations.

Prior to January 11, 2007, United States intelligence agencies detected two previous tests of the Chinese ASAT system and weeks before the *FY-1C* ASAT test U.S. intelligence agencies were aware the Chinese were preparing another ASAT test.[67] This fact raises the question: is the obligation to conduct international consultations affected or voided when States have knowledge of planned activities gained through intelligence apparatus? The answer is no. The international consultation obligation applies whether or not other States have knowledge of planned activities that they gained through intelligence apparatus. The critical point is that Article IX requires States to *consult* and does not distinguish or modify this obligation towards States that have independently gained knowledge of a planned space activity or experiment. The obligation to consult is not modified or negated even though States *may request* consultation concerning an activity or experiment they have reason to

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[66] *See* BBC NEWS REPORT, *supra* note 58.

[67] Michael Gordon & David Cloud, *U.S. Knew of China's Missile Test, but Kept Silent*, NEW YORK TIMES (Apr. 23, 2007).

believe would cause potentially harmful inference with activities in the peaceful exploration and use of outer space.

China's violation of the international consultation obligation must be considered in light of State practice. During the Cold War, The United States and Soviet Union experimented with various ASAT weapons. ASAT experiments against orbiting satellites were successfully carried out by the United States and Soviet Union. The last successful kinetic ASAT experiment was carried out by the United States in 1985 on the *Solwind* satellite.[68] Neither the United States nor the Soviet Union conducted international consultations in accordance with Article IX prior to conducting their kinetic ASAT activities. The Chinese failure to undertake appropriate international consultations prior to conducting their ASAT tests is consistent with the Cold War practices of the United States and Soviet Union.

Do the Cold War practices of the United States and Soviet Union establish an agreement of State Parties to the Treaty to interpret Article IX as not requiring appropriate international consultations prior to conducting kinetic ASAT activities or experiments in outer space? Article 31(3)(B) of the Vienna Convention states: "[t]here shall be taken into account, together with the context...any subsequent practice in the application of the treaty which establishes the agreement of the parties regarding its interpretation." "It is not necessary to show that each party has engaged in a practice, only that all have accepted it, albeit tacitly."[69]

Let us assume that the U.S. and Soviet kinetic ASAT activities did trigger the obligation to undertake appropriate international consultation. In that case, one can argue that the Cold War practice of the United States and Soviet Union establishes an agreement among States Party to the Treaty to interpret Article IX as not requiring appropriate international consultations prior to conducting kinetic ASAT activities or experiments in outer space. In support of this argument, one can present a lack of objection to the practice as tacit approval by other States

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[68] *Judge Won't Bar Test Firing at Satellite, Expected Today*, NEW YORK TIMES A8 (Sept. 13, 1985).

[69] AUST, *supra* note 11, at 243.

Party to the Treaty. One can also argue that when taking into account subsequent State practice, more weight should be given to the subsequent practice of States that have actually conducted kinetic ASAT experiments in outer space. To date, only three States (the United States, the former Soviet Union, and China) are known to have conducted such experiments and none of these States presumably undertook appropriate international consultations prior to conducting kinetic ASAT activities or experiments in outer space. It can be argued that the subsequent practices of these three States establish an agreement among Parties that excludes kinetic ASAT tests from Article IX's obligation to conduct appropriate international. It can be further argument that all have accepted it by tacit approval.

While these arguments have merit, they fail to overturn the presumption that States intend to be bound by the terms of their written agreements in accordance with the ordinary meaning to be given to the treaty in their context and in light of its object and purpose.

First, consider that subsequent practice in the application of the treaty is much more effective at establishing agreement as to treaty interpretation when the treaty is bilateral or has a limited number of State Parties and that the Outer Space Treaty has almost half of the world's nations as States Party. While three States have presumably failed to conform to Treaty obligations, no State has formally supported an exclusion of kinetic ASAT tests from Article IX international consultation obligations. Furthermore, there is uncertainty as to what States have or had the technological capacity to monitor, detect, and verify a kinetic ASAT test has occurred and by whom the test was conducted. During the Cold War these technological limitations presumably only allowed a handful of States to verify firsthand an ASAT test occurred. It is unjust to argue tacit approval by States when States did not have independent methods of ASAT test verification.

Also consider that following the *FY-1C* ASAT test, Britain, Australia, Canada, Japan, Taiwan, India, South Korea and the European Union joined the United States in protesting and call-

ing upon Beijing for consultations.[70] While the lack of objection during the Cold War is an important subsequent practice, so are the objections raised after *FY-1C*.

The reasonable conclusion is that subsequent state practice has not yet established that Article IX's appropriate international consultation obligation excludes consulting for kinetic ASAT activities or experiments in outer space. However, if States continue to perform kinetic ASAT experiments without conducting appropriate international consultations, the argument in favor of subsequent State practice establishing an agreement among States Party to exclude kinetic ASAT activities from Article IX's international consultation provision is strengthened.

ii. Obligation to Avoid Harmful Contamination of Outer Space

Did China satisfy the Article IX obligation to avoid harmful contamination of outer space?

Polar (PO) and low-earth orbits (LEO) are used for a variety of purposes. Remote sensing, manned space flight, communication satellites, the *International Space Station*, and a variety of other space objects and personnel occupy PO and LEO at any given time. The Chinese ASAT test introduced thousands of pieces of potentially hazardous space debris into PO and LEO[71] that will be in the outer space environment, in substantial amounts, for decades.[72] This space debris has modified the PO and LEO environment making orbits that intercept the Chinese ASAT debris field unfit (or at the least dangerous) for use. Given the amount of debris released, it seems reasonable for this contamination to be considered harmful or of a kind likely to be injurious.

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[70] Hitchens, *supra* note 57, at 23.

[71] *See* Kelso, *supra* note 3; *China's ASAT Test Will Intensify U.S.-Chinese Standoff*, *supra* note 2; Peter Siegal & James Gertzensang, *Chinese Missile Strikes Satellite*, THE L.A. TIMES, Jan. 19, 2007, http://www.latimes.com/news/nationworld/world/la-fg-satellite19jan19,0,3917551.story?coll=la-homeeadlines>%20and%20Technology%20 News%20Daily%20; *and* Joseph Kahn, *China Confirms Space Test; Denies Intent to Intimidate*, THE NEW YORK TIMES (Jan. 24, 2007), http://www.nytimes .com/2007/01/24/world/asia/24china.html?emc=eta1.

[72] Morring, *supra* note 64.

Article IX does not prohibit harmful contamination of outer space. Instead, states are obligated to *avoid* harmful contamination. China only violated this provision if they did not conduct the ASAT test to avoid harmful contamination. Determining whether an action satisfactorily avoided harmful contamination is difficult because the Treaty does not provide the procedure for appropriate international consultations nor designate an agency to which States should turn for an authoritative evaluation.[73]

Conducting an ASAT test on an orbiting satellite does not in and of itself constitute *unavoidable* harmful contamination of outer space. It is possible that the underlying target or technology of an ASAT test will generate unavoidable space debris. However, kinetic ASAT tests can be conducted against targets in orbits with altitudes and inclinations that would minimize harmful contamination.

It does not appear that the Chinese attempted to modify the target satellite's orbit in order to avoid harmful contamination or minimize the amount of time the resulting debris field would remain in outer space. On this basis, an argument exists that the Chinese did violate the harmful contamination provision of Article IX. However, this argument is tenuous due to the ambiguous and subjective nature of establishing a standard for *avoiding* harmful contamination. Therefore, it cannot be definitively assessed whether the Chinese ASAT test violated the harmful contamination provision of Article IX.

B. USA-193

In January 2008, the United States announced publicly that it had lost control of a satellite, *USA-193* (a.k.a. *NROL-21*), whose orbit was decaying and would eventually bring the satellite into the Earth's atmosphere.[74] On February 14, 2008, the United States Department of Defense held a news briefing publicly addressing the decay and planned kinetic ASAT intercept of *USA-193*. U.S. officials indicated they were communicating

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[73] Vlasic, *supra* note 50, at 517.

[74] Paul Harris, *US warns out-of-control spy satellite is plunging to Earth*, THE OBSERVER, Jan. 27, 2008, www.guardian.co.uk.

with other countries and various organizations (e.g. the U.N., NATO, and ESA) to inform them of the actions the U.S. planned to take regarding *USA-193*. However, the U.S. Deputy National Security Advisor stated:

> The United States has certain obligations based on treaties and other agreements related to activities in space. The 1967 U.N. Treaty on Exploration and Use of Outer Space, in particular, calls on states to keep others informed of activities of potential concern. While we do not believe that we meet the standard of Article IX of that Treaty that says we would have to consult in the case of generating potentially harmful interference with other activities in space, we do believe it is important to keep other countries informed of what is happening.[75]

This statement reveals that the United States did not believe the planned intercept of *USA-193* triggered the international consultation provision of Article IX. Was this statement legally accurate or did the planned intercept of *USA-193* trigger the international consultation provision?

The U.S. planned on intercepting *USA-193* with a kinetic ASAT missile just prior to it hitting the Earth's atmosphere.[76] The U.S. estimated that over 50 percent of debris generated from the interception would be de-orbited within the two orbits. The U.S. did consider whether unmanned bodies in space, in low-Earth orbit, and the space station would be at increased risk of space debris collisions.[77] The U.S. stated they were planning their activities with "due regard" to the safety of people in orbit.[78]

On February 21, 2008, the U.S. successfully intercepted *USA-193*. In accordance with the Outer Space Treaty and Registration Convention, the United States notified the U.N.[79] The

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[75] *See DoD News Briefing, supra* note 5.
[76] *Id.*
[77] *Id.*
[78] *Id.*
[79] *See* Outer Space Treaty, *supra* note 7, at art. XI. *See also* the Convention on the Registration of Objects Launched into Outer Space, 28 U.S.T. 695, 1023 U.N.T.S. 15.

interception occurred at an altitude of approximately 133 miles.[80]

i. International Consultation Obligation

Did the planned intercept of USA-193 trigger the international consultation provision or was the U.S. correct in asserting that their planned ASAT intercept did not meet the standard of Article IX?

As discussed above, before a State is obligated to undertake international consultations three conditions must be satisfied: (1) There is an activity or experiment in outer space (e.g. an ASAT activity or test), including the Moon and other celestial bodies, planned by the State or its nationals,[81] (2) the State must have reason to believe the activity or experiment (e.g. an ASAT activity or test) would cause potentially harmful interference,[82] and (3) that this potentially harmful interference must potentially interfere with the activities of other States Parties to the Outer Space Treaty in the peaceful exploration and use of outer space, including the Moon and other celestial bodies.[83]

It can be argued that the planned intercept of *USA-193* was not going to occur "in outer space." The failure of international law to delimitate airspace and outer space leaves some ambiguity as to whether the height of the *USA-193* intercept was in outer space. However,

> since no State has ever claimed that a satellite orbiting the Earth was infringing its national airspace, it is possible to say that in international law, outer space begins *at least* from the height above the Earth of the lowest perigee of any existing or past artificial satellite that has orbited the Earth without encountering any protest.[84]

On this basis, the intercept of *USA-193* did occur in outer space.

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[80] Kris DeRago, *Military confirms destruction of satellite*, UNIVERSITY WIRE (Feb. 25, 2008).

[81] *See* Outer Space Treaty, *supra* note 7, at art. IX.

[82] *Id.*

[83] *Id.*

[84] Christopher M. Petras, *"Space Force Alpha:" Military Use of the International Space Station and the Concept of "Peaceful Purposes"*, 53 A.F.L. REV. 135, 155 (2002).

Article IX only requires international consultations when a State has reason to believe a planned activity or experiment would cause potentially harmful interference with other States Parties in the peaceful exploration and use of outer space. *USA-193*'s intercept was designed to limit the lifetime of space debris generated from the event by conducting the intercept as *USA-193* entered the final stages of a decaying orbit. It is questionable whether the U.S. had reason to believe the planned intercept would cause potentially harmful interference because it was estimated the debris would remain in orbit a short time and that the bulk of the debris would be in an orbit not often utilized. The intercept would definitely have caused interference, but whether or not it gave reason to believe potentially harmful interference with other State activities is not conclusive.

Comparing USA-193 to *FY-1C*, while the *FY-1C* interpret would certainly generate significant space debris that would remain in orbits that are utilized by other States, the *USA-193* intercept would occur in a decaying orbit, at a low altitude, with a minimal lifetime for space debris generated.

It was concluded above that China should have had reason to believe that their planned experiment would have caused potentially harmful interference. In that analysis, China was appropriately held to the standard that when interpreting and applying Article IX Treaty obligations a State must do so in good faith and due regard to the Treaty. If one applies the principle of good faith and due regard to the *USA-193* intercept, concluding that the U.S. had "reason to believe" is with merit even though the anticipated impact of *USA-193*'s intercept would be significantly less then *FY-1C*'s.

While concluding the U.S. had "reason to believe" has merit, the United States was also correct in stating the position that "we do not believe that we meet the standard of Article IX of that Treaty that says we would have to consult in the case of generating potentially harmful interference,"[85] because this is a statement *only* of the United States own evaluation whether they have reason to believe their planned activities would cause

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[85] *See DoD News Briefing, supra* note 5.

potentially harmful interference. Solely on the basis of the U.S. evaluation, the intercept of *USA-193* was not subject to Article IX international consultation obligations. The U.S. position is defendable because the objective evidence presented prior to the planned intercept of *USA-193* does not conclusively establish whether the planned intercept would give "reason to believe" that interference would be generated that was potentially harmful to other State activities in outer space. Unlike *FY-1C*, the interception orbit of *USA-193* would not result in a long lasting debris field of significant size and any debris generated would primarily be in a low altitude decaying orbit. It was within the discretion of the United States to conclude the planned intercept of *USA-193* did not give "reason to believe" potentially harmful interference would occur because the facts did not definitely establish the U.S. should have had "reason to believe."

While this exercise of this discretion by the United States was within the bounds of Article IX, it was also arguably a strategic mistake. As discussed above, the Outer Space Treaty is primarily a treaty of principles, crafted for the purposes of *proscribing* norms to an area where uncertainty existed as to what law, if any, applied. These proscriptive principles, by their very nature, cannot embody their object and purpose by solely reviewing their text. As a result subsequent State actions play a role in the interpretation and application of the Treaty. The U.S. is able to defend its conclusion that Article IX consultation were not triggered in part because State practice has yet to clearly establish the scope of the consultation obligation.

Why was denial a mistake? Even though the U.S. denied Article IX applied to the *USA-193* intercept, the U.S. still undertook international consultations and informed the international community of their planned activity. The U.S. essentially fulfilled the minimum requirements of appropriate international consultations: to contact States Parties to the Treaty whose peaceful explorations and use of outer space would experience potentially harmful interference and provide them with information sufficient to take appropriate action to prevent potentially harmful interference with their uses or explorations in outer space, the Moon and other celestial bodies.

The U.S could have proffered that Article IX international consultation obligations were applicable. If so, the procedure and substantive nature of the consultations the United States undertook would have established a precedent of State practice with regards to Article IX. This was a unique opportunity to guide the application of Article IX. Instead, the U.S. essentially fulfilled the Article IX requirements without establishing a precedent to clarify Article IX obligations. As a result, other States planning kinetic ASAT activities and experiments will not have an historical legal precedent of a State Party recognizing and adhering to Article IX international consultation obligations. If the U.S. had recognized Article IX's application, States that plan to conduct kinetic ASAT experiments would be under greater scrutiny if they failed to recognize Article IX's application and conduct appropriate international consultations. U.S. recognition would also have set a threshold of debris generation that clarifies when a State should "have reason to believe" their planned activity or experiment in space would cause potentially harmful interference.

ii. Obligation to Avoid Harmful Contamination of Outer Space

As discussed above, conducting a kinetic ASAT intercept on an orbiting satellite does not in and of itself constitute unavoidable harmful contamination of outer space. It is possible that the underlying target or technology of an ASAT test will generate unavoidable space debris. However, kinetic ASAT tests can be conducted against targets in orbits with altitudes and inclinations that would minimize harmful contamination.

The Treaty is silent on appropriate measures or authoritative evaluations to determine whether a State has taken appropriate measures to avoid harmful contamination; however, the United States recognized *USA-193* would create harmful contamination and took measures to avoid it by intercepting the satellite in a decaying orbit at a low altitude, minimizing the lifetime of space debris generated and other potentially harmful contaminates in the outer space environment. These actions were in accordance with the principle of due regard enumerated in Article IX and the U.S. fulfilled the obligation to avoid harm-

ful contamination by taking these actions. As an example of subsequent State action, the harmful contamination and debris mitigation measures undertaken by the U.S. will contribute to interpreting the obligation of harmful contamination avoidance, as least with regards to kinetic ASAT satellite intercepts.

VI. CONCLUSIONS

China's *FY-1C* ASAT experiment violated the appropriate international consultation provisions of Article IX of the Outer Space Treaty. The orbit of *FY-1C* was of such a nature that is was reasonable to conclude prior to conducting the ASAT experiment that a successful intercept would create a debris field of size and duration that would cause potentially harmful interference with the peaceful uses and explorations of other States Party to the Treaty.

In the defense of China, their lack of consultation *prior* to the ASAT experiment is consistent with the Cold War practices of the United States and the Soviet Union (assuming the Cold War kinetic ASAT experiment of the United States and Soviet Union triggered the international consultation obligation). Nonetheless, State practice has yet to definitively establish that kinetic ASAT activities and experiments are granted an exception to Article IX obligation to conduct appropriate international consultations.

The *USA-193* intercept arguably did not violate and may not have been subject to Article IX's international consultation obligation. If the planned *USA-193* intercept did trigger the obligation to conduct appropriate international consultations, the United States met the ad minimum requirements of this obligation.

The United States denied that Article IX's international consultation obligation was applicable to the planned *USA-193* intercept. By denying its application and not recognizing its application and establishing precedent of State practice, the United States lost a strategic opportunity to shape the interpretation and development of Article IX.

The violation of Article IX's international consultation provision by China continues a disturbing trend of States not recognizing the application of this provision to kinetic ASAT activi-

ties. If this trend continues, the argument in favor of subsequent State practice establishing an agreement among States Party to exclude kinetic ASAT activities from Article IX's international consultation provision is strengthened. Fortunately several States protested China's failure to consult in the days following *FY-1C*'s destruction, reversing the trend of States' tacit approval implicit by silence.

One reason Article IX has failed to ensure States fulfill their obligations is because the Outer Space Treaty does not provide a procedure for appropriate international consultations nor designate an agency to which States should turn for an authoritative evaluation of their planned activities.[86] As a result Article IX's procedural and substantive application is largely left to the discretion of States, and States determine themselves whether the obligation to consult is triggered. In our anarchic system of international relations, States have little motivation to interpret and apply agreements to restrict their freedom of actions without assurance that other States will act in kind. This failure is a manifestation of the much greater problem: the vacuum of supranational authority in international law. Today no supranational authority exists that can rule on the legality of State action and enforce this judgment independent of State influence. As a result, the immediate self-interests of States support restrictive interpretation and application of Article IX.

Even with self-interest supporting restrictive interpretations, as discussed *supra*, an interpretation of the Outer Space Treaty based on the Vienna Convention methodology does provide some degree of objectivity for States who are assessing their planned activities. For example, States are obliged to consider their planned activities in light of the principle of due regard and interpret provisions of Article IX in good faith.

In instances of ambiguity, the Outer Space Treaty does not provide a mechanism of interpretation or dispute resolution. Fortunately, in the event of dispute arising due to interpretation and application of the Treaty States can rely on and should make use of the U.N. Charter's mechanism of pacific settle-

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[86] Vlasic, *supra* note 50.

ments of disputes.[87] When a State clearly violates Article IX, the international community should react with unified condemnation, take appropriate actions to discipline the violator (in accordance with the U.N. Charter), and ensure violations are not repeated in the future. It is critical for States to appreciate that at this point in history, State practice and application of Article IX will largely determine whether or not Article IX is rendered ineffectual. It is in the interests of all nations for Article IX to be a substantive provision and not just empty words. As the *FY-1C* experiment demonstrated, the failure of States to consult prior to conducting activities increases mistrust, raises tensions, and undermines international peace and security.

The international community should take concrete steps towards strengthening Article IX. An additional protocol to the Outer Space Treaty could be negotiated that provides a procedure and authoritative body for determining whether a planned activity warrants international consultation and whether appropriate measures have been planned to avoid harmful contamination of outer space and adverse changes to the Earth's environment. Another option is for the international community to agree to standards of harmful contamination and harmful interference mitigation independent of an additional protocol.

Concluding that Article IX imposes a substantive obligation and further concluding that the *FY-1C* intercept violated this obligation is a serious charge. While we all bring to a discussion the bias of our individual human experience, this author recognizes such bias exists and has attempted to assess the *FY-1C* and *USA-193* intercepts objectively. There is no doubt that all people, regardless of nationality, share the common human desires of self-preservation and peace. It is therefore with hope that these conclusions are made. The hope that States, when given greater clarity as to the scope of Article IX's legal obligations, will more fully respect and carry out their respective agreements.

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[87] *See* U.N. Charter arts. 33-38.

INTERNATIONAL SPACE EXPLORATION AND CRITICAL TRANSPARENCY OF *BASIC* RESEARCH: IMPACT OF THE U.S. INTERNATIONAL TRAFFIC IN ARMS REGULATIONS

George S. Robinson[*] &
Eric McAdamis[**]

I. INTRODUCTION

In January 2004, President George W. Bush announced a Vision for Space Exploration (VSE) that, in part, encouraged and mandated for the United States the pursuit of "opportunities for international participation to support U.S. space exploration goals."[1] In recent years, however, U.S. scientists have expressed a significant and growing frustration with their inability to collaborate and exchange information and research data effectively with foreign colleagues in pursuing that mandate for space exploration. The source of much of this frustration has been the confusion caused by inconsistencies in the interpretation and application of pivotal terms, such as "basic research," "fundamental research," "applied research," and "public domain" as these terms appear primarily in the U.S. Interna-

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[*] Dr. Robinson, retired from the Smithsonian Institution, is currently in private law practice, and serves on several boards of trustees and advisory committees, including the National Aeronautics and Space Administration's Planetary Protection Subcommittee of the NASA Advisory Council. He received an AB from Bowdoin College ('60), an LL.B from the University of Virginia ('63), an LL.M. from the McGill University Institute of Air and Space Law ('67), and the first Doctor of Civil Laws degree from McGill University's Graduate Law Faculty, Institute of Air and Space Law ('71).

[**] Eric McAdamis is a recent graduate of The University of Mississippi School of Law where he served as an intern for the National Center for Remote Sensing, Air and Space Law. He is currently preparing to perform volunteer legal work for the Commonwealth Human Rights Initiative in Accra, Ghana and then plans to begin work on a Ph.D. in judicial politics in the fall of 2009.

[1] *See* Press Release, The Whitehouse, President Bush Announces New Vision for Space Exploration Program (January 14, 2008), http://www.whitehouse.gov/news/releases/2004/01/20040114-3.html.

tional Traffic in Arms Regulations (hereinafter the ITAR) and enabling documents.[2] These regulations were intended and drafted principally to protect the economic, military, and diplomatic interests of the United States. History has shown in large part that, in many instances, these regulations and their applications are self-defeating with respect to those goals.

While individuals involved with research in the space sciences recognize the necessity of treating certain hardware and technology as militarily sensitive, they also believe that the ITAR are much too inclusive; and unnecessarily so. Further, the United States has many policy priorities in space other than national security, including its diminishing leadership role in space exploration, global commercial competitiveness, re-establishing university excellence in the space-related sciences and technologies, and encouraging and maintaining critically necessary international partnerships in conducting "fundamental research" in space and space-related matters. There are ongoing costs and delays of significance in the processing of ITAR requirements, particularly as they relate to basic research proposals relying in part on foreign colleagues, foreign nations, and foreign funding for their undertaking. Critical international collaboration for basic and applied research involving U.S. personnel and facilities is diminishing rapidly.

Much of the confusion seems to stem from the inability or political unwillingness of the U.S. Congress and the U.S. Department of State to formulate realistic policies dealing with international relations consistent with national, regional, and global defense and security realities, and the role of space research and exploration in formulating those policies. This, combined with what seem to be intentional regulatory drafting ambiguities, has created an environment of debilitating confusion for scientists and engineers involved in U.S. civilian and shared

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[2] Arms Export Control Act, 22 USC § 2778 (1979), Priv. L. No. 96-72m, 93 Stat. 503 (1979), 22 C.F.R. §§ 120-130 (2002). The ITAR currently are administered by the Directorate of Defense Trade Control (DDTC) at the U.S. Department of State.

civilian/military space activities mandated by the VSE, consistent with the 2006 U.S. National Space Policy. [3]

In addition to inconsistencies in interpretations and applications of pivotal ITAR words and phrases, the confusion and frustration of scientists and university and non-profit research laboratory export control officers also find their roots in the increasing indifference and lack of true understanding of legislators and policymakers regarding the *absolute critical importance of maintaining and enhancing ongoing "basic research;"* add to that, also, their increasing deference to directed or "applied" research with comparatively short-term public benefits. In large part, the critical need for basic research is lost in the broadly cast net of the ITAR, i.e., the resultant decreasing transparency of *basic* scientific research as it relates to (1) traditional and crucial open collaboration among colleagues and the necessary written and oral exchanges of data/information deriving from their basic research efforts, and (2) the preservation of the serendipitous and uniquely beneficial potentials resulting from such research.

There is a pressing need to establish more refined and consistent definitions of pivotal words and phrases, such as "*basic* research," "*fundamental* research," "*applied* research," and "public domain" as used in applicable laws in order to establish continuity and consistency, both in domestic and international understandings and uses of those words and phrases in the ITAR. The observation is based in large part, although certainly not solely, on the fact that scientists and their colleagues may be held *personally* accountable for violations of the ITAR civil and criminal provisions, and their respective sanctions or penalties.

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[3] For a general commentary by the author on the 2006 U.S. National Space Policy, as well as the full text of the unclassified version of the policy, *see* G.S. Robinson, *The U.S. National Space Policy: Pushing the Limits of Space Treaties?*, ZEITSCHRIFT FÜR LUFT- UND WELTRAUMRECHT [GERMAN J. OF AIR & SPACE L.], 45-57 (ZLW 56.Jg. 1/2007).

II. BACKGROUND

Since at least the late 1930s, the United States has restricted in one fashion or other the export of certain goods and technology. The objective, of course, is protection of its national security and foreign policy interests by maintaining control over the export of certain goods, technologies, and services that might be used in military development of other nations; particularly those that may have hostile intentions toward the United States and its allies. At present, three governmental Departments are involved in promulgating a variety of regulations aimed at securing the nation's various interests, including those relating to national security and economic competitiveness. The Department of Commerce regulates "dual use" items through its Export Administration Regulations (EAR), the Department of State through its International Traffic in Arms Regulations (ITAR),[4] and the Department of Treasury through its designation of trade embargoes by its Office of Foreign Assets Control.[5]

The ITAR assist in controlling trade in defense items and services as stated by the Arms Export Control Act, and Executive Order 11958, as amended. All categories on the U.S. Munitions List (USML) set forth the defense items and articles to be regulated by the ITAR.[6] These regulations also control all space satellites that were placed on the USML by the Strom Thurmond National Defense Authorization Act of FY 1999,[7] and Category XV of the USML includes "Spacecraft Systems and

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[4] EAR regulations are found at 15 C.F.R., Chap. VII, Part 734. The ITAR control defense trade as referenced in 22 U.S.C. § 2788 of the Arms Export Control Act [*see* 22 U.S.C. § 2778 (1979), Priv. L. No. 96-72, 93 Stat. 503 (1979), 22 C.F.R §§ 120-130 (2002)].

[5] 31 C.F.R. Part 500. The Office of Foreign Assets Control (OFAC) "administers and enforces economic and trade sanctions based on U.S. foreign policy and national security goals against targeted foreign countries and regimes, terrorists, international narcotics traffickers, those engaged in activities related to the proliferation of weapons of mass destruction." Office of Foreign Assets Control, *Mission,* http://www.ustreas.gov/offices/enforcement/ofac/ (last viewed Nov. 3, 2008). In the context of activities requiring collaborative efforts with member states of the European Union (EU),that organization has condemned, along with other entities, certain OFAC and related security laws aimed at various embargoed nations.

[6] The United States Munitions List, 22 C.F.R. § 121.1 – 121.16 (2008).

[7] Strom Thurmond National Defense Authorization Act of FY 1999, Pub. L. No. 105-261, 112 Stat. 1920 (1998).